Lab Practice Problem 10B

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In this activity, you will practice converting numbers between different number systems. You will apply several common conversion techniques and use shortcuts when appropriate. The goal of this exercise is to strengthen your understanding of positional notation and how values are represented across number systems.

General Instructions

Read through each conversion technique carefully and review the worked examples. Recall that decimal is base 10, binary is base 2, and hexadecimal is base 16, where the base represents the number of symbols used in the number system. For each problem, use the conversion method that is most appropriate for the given number bases. Use the following table as a reference when working with hexadecimal values.

Decimal	Binary	Hexadecimal
0	0000	0
1	0001	1
2	0010	2
3	0011	3
4	0100	4
5	0101	5
6	0110	6
7	0111	7
8	1000	8
9	1001	9
10	1010	A
11	1011	B
12	1100	C
13	1101	D
14	1110	E
15	1111	F

1. Any Base to Decimal

To convert a number from any base to decimal, use the following method:

$$(d_n d_{n-1} \cdots d_1 d_0)_b = d_n b^n + d_{n-1} b^{n-1} + \cdots + d_1 b^1 + d_0 b^0$$

Example 1

Convert $(1011)_2$ to decimal.

$$(1011)_2 = 1(2^3) + 0(2^2) + 1(2^1) + 1(2^0)$$

$$= 8 + 0 + 2 + 1 = 11$$

$$(1011)_2 = (11)_{10}$$

Example 2

Convert $(2F)_{16}$ to decimal.

$$(2F)_{16} = 2(16^1) + 15(16^0)$$

$$= 32 + 15 = 47$$

$$(2F)_{16} = (47)_{10}$$

2. Decimal to Any Base

To convert a decimal number to another base, use the division-remainder technique.

Important: The remainders must be read in reverse order.

Example 1

Convert $(13)_{10}$ to binary.

$$13 \div 2 = 6 \text{ remainder } 1$$

$$6 \div 2 = 3 \text{ remainder } 0$$

$$3 \div 2 = 1 \text{ remainder } 1$$

$$1 \div 2 = 0 \text{ remainder } 1$$

Reading the remainders in reverse:

$$(13)_{10} = (1101)_2$$

Example 2

Convert $(45)_{10}$ to hexadecimal.

$$45 \div 16 = 2 \text{ remainder } 13$$

$$2 \div 16 = 0 \text{ remainder } 2$$

Since remainder 13 is `D` in hexadecimal:

$$(45)_{10} = (2D)_{16}$$

3. Any Base to Another Base (Non-Decimal to Non-Decimal)

One common technique is to convert to a common base first, such as base 10.

$$\text{Base A} \rightarrow \text{Decimal} \rightarrow \text{Base B}$$

Example 1

Convert $(11010)_2$ to octal (base 8).

Step 1: Convert to decimal

$$(11010)_2 = 1(2^4) + 1(2^3) + 0(2^2) + 1(2^1) + 0(2^0)$$

$$= 16 + 8 + 2 = 26$$

Step 2: Convert decimal to octal

$$26 \div 8 = 3 \text{ remainder } 2$$

$$3 \div 8 = 0 \text{ remainder } 3$$

$$(11010)_2 = (32)_8$$

Example 2

Convert $(57)_8$ to hexadecimal.

Step 1: Convert to decimal

$$(57)_8 = 5(8^1) + 7(8^0) = 40 + 7 = 47$$

Step 2: Convert decimal to hexadecimal

$$47 \div 16 = 2 \text{ remainder } 15$$

Since 15 is `F` in hexadecimal:

$$(57)_8 = (2F)_{16}$$

4. Shortcut Between Bases That Are Powers of 2

For bases that are perfect powers of 2, such as binary, octal, and hexadecimal, you can often use a shortcut instead of converting through decimal.

Important Note

Octal (base 8): group bits into sets of **3**
Hexadecimal (base 16): group bits into sets of **4**

Example 1: Binary to Octal

Convert $(101110)_2$ to octal.

Group into sets of 3 from right to left:

$$101\ 110$$

$$101_2 = 5, \quad 110_2 = 6$$

$$(101110)_2 = (56)_8$$

Example 2: Binary to Hexadecimal

Convert $(11010111)_2$ to hexadecimal.

Group into sets of 4 from right to left:

$$1101\ 0111$$

$$1101_2 = D, \quad 0111_2 = 7$$

$$(11010111)_2 = (D7)_{16}$$

Example 3: Hexadecimal to Binary

Convert $(A3)_{16}$ to binary.

$$A = 1010, \quad 3 = 0011$$

$$(A3)_{16} = (10100011)_2$$

Example 4: Octal to Binary

Convert $(57)_8$ to binary.

$$5 = 101, \quad 7 = 111$$

$$(57)_8 = (101111)_2$$

Try It Out

Important: You must show the conversion steps used and not simply write the final answer. You are also not allowed to use a calculator for this activity.

A. Convert to Decimal

1. $(2AE)_{16} = \underline{\hspace{4cm}}_{10}$

2. $(127)_8 = \underline{\hspace{4cm}}_{10}$

B. Convert from Decimal

3. $(53)_{10} = \underline{\hspace{4cm}}_{2}$

4. $(487)_{10} = \underline{\hspace{4cm}}_{16}$

C. Convert Between Bases

5. $(52)_6 =$ _______________$_9$

6. $(213)_7 =$ _______________$_5$

7. $(723)_8 =$ _______________$_{16}$

8. $(C0DE)_{16} =$ _______________$_8$